INCH-POUND

MIL-V-70579B (AR)
27 August 1990
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

VOLTAGE CONVERTER: 9360839

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the voltage converter used on Gun, Anti-Aircraft Artillery, Self-Propelled, 20 MM, M163A2 and Towed 20 MM, M167A2. The voltage converter shall convert nominal 28 Vdc input to five isolated dc outputs and one isolated ac output. The voltage converter shall provide regulation against variations in input voltage and output current, and shall incorporate internal circuitry to protect against output overloads and internal over voltages (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-6051 - Electromagnetic Compatibility Requirements, Systems

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-S-19500/507	-	Semiconductor Device, Diode, Bipolar Transient Suppressor Type
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-70580	-	Circuit Board, 400 Hz Inverter, Voltage Converter: 9397383
MIL-C-70581	-	Circuit Board, +5 Volt Converter, Voltage Converter: 9397383
MIL-C-70582	-	Circuit Board, ±24 Volt Converter, Voltage Converter: 9397385
MIL-C-70583	-	Circuit Board, ±15 Volt Converter Voltage Converter: 9397384
MIL-C-70584	-	Circuit Board, Control and Synchronizer, Voltage Converter: 9397382

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electro-magnetic Interference
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360839	-	Voltage Converter Assembly
9397409	-	Voltage Converter Schematic

PACKAGING DATA SHEET

PS9360839	-	Packaging of Voltage Converter Assembly

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360839, all associated drawings, and with all requirements specified in applicable specifications and standards. Subassemblies shall have met the requirements of the following specifications:

Subassembly		Specification
Control and Synchronizer CCA:	9397382	MIL-C-70584
+5V Converter CCA:	9397383	MIL-C-70581
±15V Converter CCA:	9397384	MIL-C-70583
±24V Converter CCA:	9397385	MIL-C-70582
400 Hz Inverter CCA:	9397386	MIL-C-70580

Where applicable, the voltage converter shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

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3.3 Environmental service conditions.

3.3.1 Vibration. The voltage converter shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of the vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The voltage converter shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 40 g's ±10%, 11 milliseconds ±10%, half-sine wave applied in each direction of 3 mutually perpendicular axes. The G value of each impulse shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5 1/2 ms.

3.3.3 Altitude (Non-operating). The voltage converter shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The voltage converter shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

3.3.4.2 Non-operating (storage). The voltage converter shall not suffer permanent damage when subjected to the constant tempera ture extremes of -60 deg F for a period of 24 hours and +160 deg F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The voltage converter shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. The electrical characteristics of the voltage converter shall be as specified on the voltage converter schematic diagram 9397409.

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3.5 Electromagnetic interference. The voltage converter assembly shall comply with the following class 3A requirements of MIL-STD-461C:

CE03 on 28 volt input power leads
CS06 on 28 volt input power leads
RE02
RS03

The voltage converter assembly shall introduce no intrasystem electromagnetic incompatibilities.

3.6 Performance.

3.6.1 Functional description. The voltage converter shall provide regulation against variations in input voltage and output current. The voltage converter shall contain 5 modules which protect against output overloads and internal overvoltages. The design of the voltage converter shall be in accordance with drawing 9360839.

3.6.2 Power supply. The voltage converter shall operate from a battery source which varies between 22 and 30 volts dc. Battery return and chassis ground are common external to the voltage converter. The maximum current under full load conditions shall be 15 amperes (A). The voltage converter shall automatically de-energize when the input voltage falls below 16.45.

3.6.3 Input transient protection. The voltage converter shall not experience damage when subjected to transient inputs as defined in MIL-STD-704, Figure 10. The voltage converter shall contain a transzorb, J1N6056A per MIL-S-19500/507, or equivalent, across the input to protect against transients up to 200 V peak.

3.6.4 Reverse polarity. The positive (+) side of the 24 Vdc battery input, and all outputs from the voltage converter shall be isolated from chassis ground. The negative (-) side of battery input shall be tied to frame ground. A reverse polarity, 30 volts maximum, impressed on the input shall not damage the voltage converter.

3.6.5 Isolation voltage. The minimum isolation voltage shall be 50 Vdc when measured between input and outputs, input and case, and outputs and case for dc voltages, and 200 Vrms for ac voltage output to input, and output to case.

3.6.6 Insulation resistance. Insulation resistance shall be 50 megohms minimum between input and outputs, input and case, and outputs and case when measured at 50 Vdc.

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3.6.7 Dc output regulation. The outputs shall not deviate from their nominal value by more than the amount specified in Table I as a result of any combination of the following:

a. Input variations in accordance with 3.6.2.
b. Output current variations per Table I
c. Temperature variation from -40 deg F to +160 deg F
d. Initial setpoint tolerance.

TABLE I. Dc outputs

OUTPUT VOLTAGE (V)	OUTPUT CURRENT (A)		OUTPUT REGULATION	OUTPUT RIPPLE (mV rms)	OVERVOLTAGE LIMITS (V)	OVER CURRENT LIMITS (A)	
	MIN	MAX				MIN	MAX
+5.2	7.5	15	±3%	20	7.0 max	17.5	22.0
+15	0.3	1.6	±3%	20	18.0 max	2.7	7.0
-15	0.3	1.6	±3%	20	-18.0 max	2.7	7.0
+24	0.5	1.0 2.0pk*	±5%	20	28.0 max	2.7	7.0
-24	0.5	1.0 2.0pk*	±5%	20	-28.0 max	2.7	7.0

* Capable of infrequent unidirectional current pulses of 2 amps for 100 milliseconds.

3.6.8 Over current protection. The dc circuitry shall not experience damage when continuously subjected to output overload current conditions and short circuit conditions. Current limiting shall commence within the range specified in Table I. The voltage converter shall recover to normal operation automatically after removal of the overload.

3.6.9 Overvoltage protection. The voltage converter shall incorporate protection circuitry to prevent the dc outputs from exceeding the overvoltage limits of Table I. Recovery from overvoltage shutdown shall be by recycling input power.

3.6.10 ±15 V, ±24V shutdown. Shutdown of either the +15 Vdc or -15 Vdc due to short circuit or overvoltage of either shall cause shutdown of the opposite polarity voltage. This requirement also applies to the ±24 Vdc.

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3.6.11 Turn-on/turn-off overshoot. There shall be no overshoot during the turn-on into loads greater than half-load. There shall be no turn-off overshoot. The overshoot limit shall be the overvoltage limits specified in Table I.

3.6.12 Inverter frequency. The inverter operating frequency for all dc to dc converters shall be between 40 and 100 KHz, which shall remain constant to ±10 percent.

3.6.13 Ac output regulation. The ac output shall remain within the limits specified in Table II for any combination of the following:

a. Input variations in accordance with 3.4.1 and 3.4.2
b. Output load variations from 25 to 50 VA
c. Temperature variation from -40 deg F to +160 deg F
d. Initial setpoint tolerance.

TABLE II. Ac output

OUTPUT VOLTAGE RANGE	FREQUENCY RANGE	WAVESHAPE AND DISTORTION
115 Vrms ±5%	400 Hz ±0.1 %	Sine, 10% max

3.6.14 Pins. The voltage converter pin assignments are specified in Table III and Table IV.

TABLE III. J1 connector

Pin	Function
H	Chassis
J	IN RET
K	+28V IN
L	+28V IN
M	IN RET

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TABLE IV. J2 connector

Pin	Function
A	+15V
B	-15V
D	+5V
E	+5V
F	5V RET
H	+24V
J	-24V
K	115V,400Hz
M	115V,400Hz
N	15 RET
P	5V RET
R	24V RET

3.7 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.8 Burn-in. The voltage converter shall meet the requirements of 3.5 after each stabilization during the following burn-in stress application: the unit shall be cycled between -40 deg C, -5 deg C, and 71 deg C +5 deg C at a rate of 6 deg C/minute ±deg C/minute with temperature stabilization at each extreme, for a total of six (6) 180 minute cycles. The last three (3) cycles shall be failure free.

3.9 Soldering. The voltage converter shall be soldered per the general requirements and task F of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections

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set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three voltage converters for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table V (Functional Test) and Table VI (Special Sampling) herein and to any or all requirements of the applicable drawings.

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4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table V shall be performed on a 100% basis.

TABLE V. Functional tests
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
103. Burn-in	3.7	4.9
Minor:	None	N/A

4.5 Special sampling. One voltage converter out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table V and shall then meet the tests in Table VI. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

TABLE VI. Special sampling

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
104. Vibration	3.3.1	4.7.1.1
105. Shock	3.3.2	4.7.1.2
106. Altitude	3.3.3	4.7.1.3
107. Temperature	3.3.4	4.7.1.4
108. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table VII (Test equipment list) or equivalent meeting minimum requirements, shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE VII. Test equipment list

ITEM NUMBER	DESCRIPTION
1. VC test fixture	Provides interface for the voltage converter input and output connectors. Contains loads and switches to monitor the voltage converter output voltages.
2. True RMS Voltmeter	Input range from 0V to 200VRMS, 1000 VRMS peak. Input impedance for 2 mV to 200 VRMS is 10 megohms minimum. Frequency range from 2 Hz to 500 Hz. Minimum resolution of 0.5 percent.
3. dc power supply	0 to 30V, adjustable. Load regulation of 1 percent or better from 0 to 15 amperes. Less than 1 percent ripple.
4. dc power supply	0 to 15V, adjustable. Load regulation of 1 percent or better from 0 to 4 amperes. Less than 1 percent ripple.
5 dc Hipot	An instrument capable of measuring isolation voltages up to 50 Vdc between input and output circuitry and the chassis.
6. ac Hipot	An instrument capable of measuring isolation voltages up to 200 VRMS between input and output circuitry and the chassis.
7. Digital Voltmeter (DVM)	Accuracy and resolution of $\pm$0.2 percent from 1 to 100 Vdc and $\pm$0.1 percent from 1 to 200 Vac, 400 Hz. Input impedance 10 megohms or greater. 300 KHz bandwidth minimum.

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TABLE VII. Test equipment list (cont'd.)

ITEM NUMBER	DESCRIPTION
8. Megohmmeter	Meter capable of measuring insulation resistance up to 100 megs at 50 Vdc between input and output circuitry and the chassis.
9. Module test fixture	Provides interface for the module connectors. Contains loads and switches necessary to test the modules.
10. Ohmmeter	0 to 1000 ohms. Accuracy and resolution of 5 percent or better. Input impedance 10K ohms minimum.
11. Distortion Analyzer	Signal analyzer capable of measuring total distortion to 0.1% full scale between 5 Hz and 600 KHz. Input impedance, distortion mode 1M ohms ±5% shunted by less than 70 pF. Voltage range 300 micro volts to 300V rms, 10 dB. Output of 0.1 ±.01V rms open circuit and output impedance of 2K ohms minimum.
12. Burn-in Load Fixture	Provides full load burn-in for the unit under test.
13. Datalogger	Stripchart recorder capable of recording time, date, temperature (-40 degrees F to +170 degrees F), ac voltages (0 to 120 Vac) and dc voltages (0 to 50 Vdc) including input, + and -15 Vdc, + and -24 Vdc and +5 Vdc.
14. Temperature Chamber/ Heat Sink	Provides proper environment for conducting burn-in test.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The voltage converter shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the voltage converter shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz. After the application of all the shock impulses, the voltage converter shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (non-operating). This test shall be performed as specified in Procedure III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4. The electrical requirements of the voltage converter shall be tested in accordance with the following, using the test equipment listed in Table VII.

4.7.3 Electromagnetic interference. The EMI requirements of 3.5 shall be tested in accordance with the requirements of MIL-STD-462. Compatibility testing, when applicable, shall be in accordance with the provisions of MIL-E-6051.

4.7.4 Performance. The voltage converter shall be tested in accordance with the following, using the test equipment listed in Table VI.

4.7.4.1 Input transient protection. The voltage converter shall be tested to ensure that the input transient protection is in accordance with the requirements of 3.5.3.

4.7.4.2 Low voltage protection. The voltage converter shall be tested to ensure that shutdown occurs during the low voltage conditions specified in 3.6.2.

4.7.4.3 Reverse polarity. The voltage converter shall be tested to ensure that the reverse polarity is in accordance with the requirements of 3.5.4.

4.7.4.4 Isolation voltage. The voltage converter shall be tested to ensure that the isolation voltage is in accordance with the requirements of 3.5.5.

4.7.4.5 Insulation resistance. The voltage converter shall be tested to ensure that the insulation resistance is in accordance with the requirements of 3.5.6.

4.7.4.6 Dc output regulation. The voltage converter shall be tested to ensure that each dc output, adjusted to its maximum current output (see Table I), is in accordance with the specified output ripple limits (see Table I) and the output tolerance limits (see Table VIII).

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TABLE VIII. Output tolerances

Output (volts)	Tolerance (volts)
+5.2	+5.04 to +5.35
+15	+14.55 to +15.45
-15	-14.55 to -15.45
+24	+22.80 to +25.20
-24	-22.80 to -25.20

4.7.4.7 Over current protection. The voltage converter shall be tested to ensure that the over current protection is in accordance with the specified requirements of 3.6.8.

4.7.4.8 Overvoltage protection. The voltage converter shall be tested to ensure that when the voltage across each dc output rises above the maximum regulation limit but no higher than the overvoltage limit in Table I, the unit will shut down. After shut down, verify that the voltage converter is not supplying the dc output voltage. Recycle the input power and verify that the dc output voltage has recovered to normal.

4.7.4.9 +/- 15V, +/- 24V shutdown. The voltage converter shall be tested to ensure that the ±15V, ±24V shutdown is in accordance with the specified requirements of 3.6.10.

4.7.4.10 Turn-on/turn-off overshoot. The voltage converter shall be tested to ensure that when the unit is turned ON, each dc output does not exceed the maximum overvoltage limits (see Table I). When the unit is turned OFF, each dc output shall not exceed the maximum overvoltage limits and shall return to 0 Vdc.

4.7.4.11 Inverter frequency. The voltage converter shall be tested to ensure that the inverter frequency of each dc to dc converter is in accordance with the specified requirements of 3.6.12.

4.7.4.12 Ac output regulation. The voltage converter shall be tested to ensure that the ac output regulation is in accordance with the specified requirements of Table II, when subjected to the variations and tolerances specified in 3.6.13.

4.8 Soldering. Soldering processes and workmanship shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified by paragraph 3.9 herein.

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4.9 Burn-in. The voltage converter shall be temperature cycled and burned-in, as required by paragraph 3.7 herein, for 6 complete cycles as shown in Figure 1, for a total of 18 hours minimum, using the burn-in test set-up shown in Figure 2. Temperature shall be measured at the fixture's heatsink.

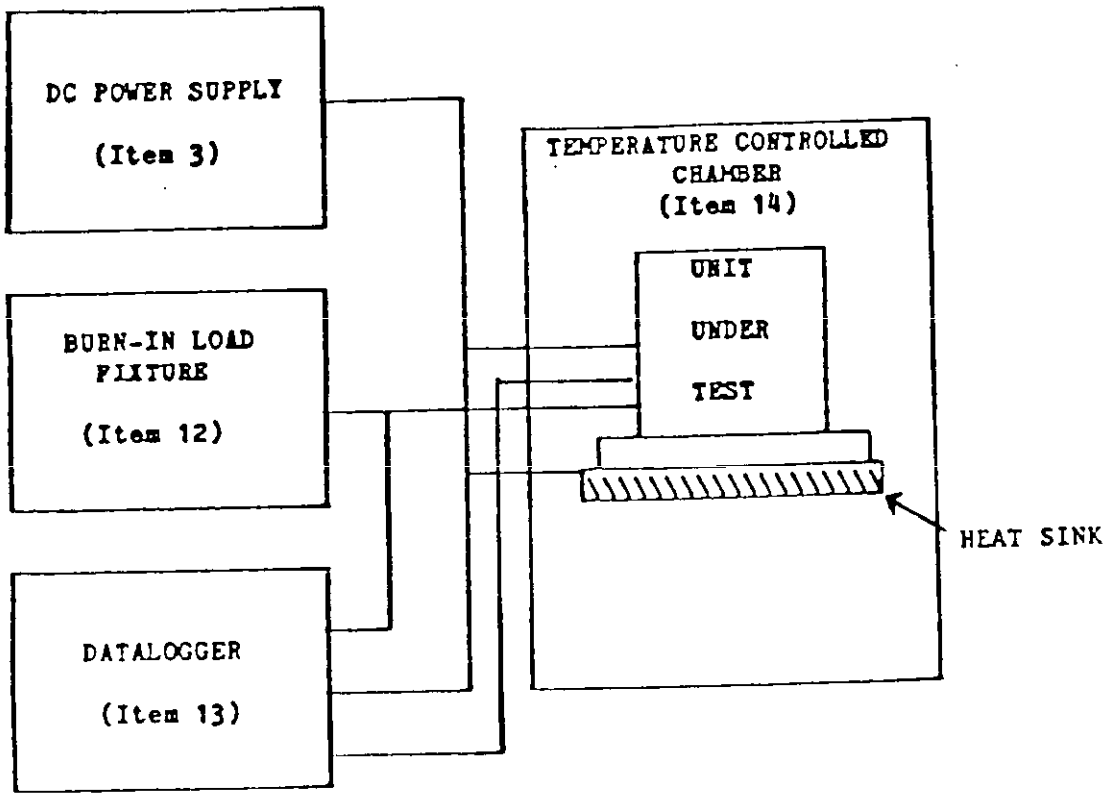
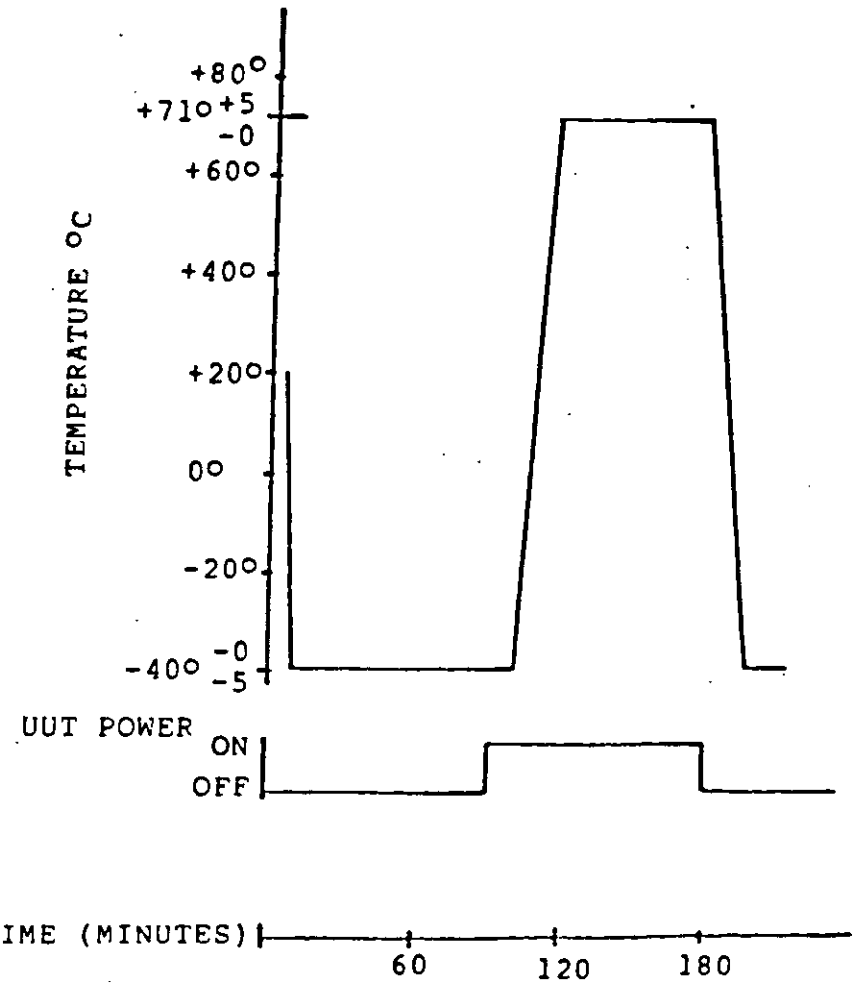


Figure 1. Burn-in Temperature Versus Time Diagram.

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NOTES:

1. Temperature rate change, 6°C/Min.

2. Dwell at temperature until unit stabilizes. Total lapsed time per cycle is calculated to be about three hours at full load.

Figure 2. Burn-in Test Set-up.

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5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS9360839.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The voltage converter is used in the PIVADS Self-Propelled, M163A2 and Towed, M167A2, vehicles.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Gun, Anti-aircraft Artillery, Self-propelled, 20 MM, M/63A2 Gun, Anti-aircraft Artillery, Towed, 20 MM, M167A2
Product Improved Vulcan Air Defence System (PIVADS)
Voltage converter

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6.6 Changes from previous issue. The previous issue were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms center (NPFC) for publication. This issue is initial release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A725)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-V-70579B (AR)	27 August 1990

3. DOCUMENT TITLE

VOLTAGE CONVERTER: 9360839

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME	b. TELEPHONE (Include Area Code)	
	(1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6674	880-6674
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